level, a *norm* usually gives an idea of the magnitude of a matrix. In a similar way, the *condition number* of a matrix gives an idea of how close the matrix is to being singular. The condition number can vary between 1 and infinity. A matrix with a large condition number is nearly singular, while a matrix with a condition number close to 1 is far from being singular.

If you have read the first part of this article, when we calculated the inverse of a singular matrix, we got back *rcond = 0*. rcond is the reciprocal of the condition number. Since the matrix was singular, its condition number is infinity and hence its reciprocal is 0.

In Octave, cond(a) is defined as norm(a, p) * norm(inv(a), p).

In a linear system of equations, the condition number of the coefficient matrix is usually an indication of whether the system of equations is well-conditioned or ill-conditioned, i.e., it is an indicator of how sensitive the solution is to the change in coefficients. In a well conditioned system, the sensitivity is ideally low.

Rank: The rank of matrix A is the maximum number of linearly independent columns in the matrix. In a matrix, the number of linearly independent columns can be shown to be equal to the number of linearly independent rows. Thus, the rank of a matrix can never be greater than the smaller dimension of the matrix. In Octave, you would use the command rank(A) to find the rank of A.

Solving a simple linear equation

Consider a system of two equations in two variables:

- $2x + 3y = 3$
- $3x + 2y = 5$

I can rewrite the above system of equations as follows:

$$\begin{bmatrix} 2 & 3 \\ 3 & 2 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 3 \\ 5 \end{bmatrix}$$

This is of the form: $Ax = B$. The solution to the above system of equations would thus be the matrix x, which satisfies the equation above. Let us seek x. Since A is a non-singular square matrix, we can write x as follows: $x = A^{-1} B$

So, now we shall tell Octave to find the inverse of A and then find the product of B with it. The result will give us x.

```
octave:6> A= [ 2 , 3 ; 3 , 2 ]
A =

2  3
3  2


octave:7> B= [ 3 ; 5 ]
B =

3
5

octave:8> [ a , rcond ] = inverse (A)
a =

-0.40000  0.60000
0.60000  -0.40000

rcond = 0.20000

octave:10 > x=a*b
x =

1.80000
-0.20000

octave:11 >
```

So, we have our x matrix and thus the solution to the given system of equations is x = 1.8, y = −0.2

Inverse function: The inverse function computes the inverse of a square matrix. It also returns an estimate of the reciprocal condition number, which we have seen earlier.

Left-division operator: The calculation of the inverse of a square matrix is an expensive process and it is rarely calculated explicitly when solving a system of equations. Matrix transformations are usually used to solve a system of linear equations without explicitly calculating the inverse. Octave has the left-division operator that can find the solution to a system of equations without requiring the explicit calculation of the inverse. For example, the above system of equations could be easily solved by the following commands:

```
octave:12 > x = A \ B
x =

1.80000
-0.20000
```

$A \backslash B$ is conceptually equivalent to the statement *inverse(A) * B*, but it is computed without explicitly forming the inverse of A.

Solving a simple linear optimisation problem

In this section, I will introduce you to the features in Octave, which can be used to solve optimisation problems. Octave has support for linear programming, quadratic programming, nonlinear programming, and linear least squares minimisation. I will now walk you through solving a simple linear programming (LP) problem.

Any LP problem can be expressed in the following standard form:

Minimize $f(x) = c^T x$
s.t. $Ax = b,\ x \geq 0$
where $b \geq 0$

Consider this simple LP problem:

Minimize $f(x) = -2x_1 - x_2$
s.t. $2x_1 - x_2 = 8$
$x_1 + 2x_2 = 14$
$x_1, x_2 \geq 0$

We will now form the matrices c, x, A, b so as to write the given problem in the standard LP form:

$$c = \begin{bmatrix} -2 \\ -1 \end{bmatrix} x = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} A = \begin{bmatrix} 2 & -1 \\ 1 & 2 \end{bmatrix} b = \begin{bmatrix} 8 \\ 14 \end{bmatrix}$$

You will notice that we can easily write the given problem in the